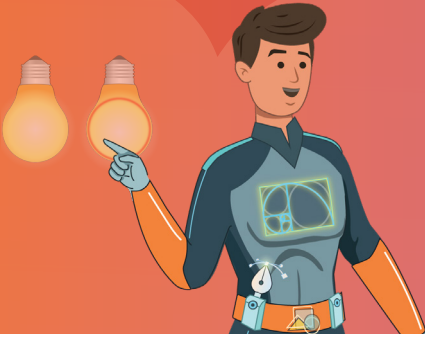


Objects Around Us - Part 1



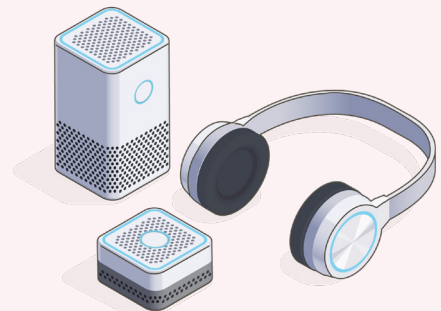
Hey DeZain! I have been playing a 3D endless runner game since yesterday. Everything looks so real that I feel I exist within the game. It feels like magic!

That's awesome! Did you know that asset creation starts with the designing of basic shapes using polygons, vertices, and edges? 3D designers add textures and materials to give the model different properties like metal, wood, or glass.



3D Design

'3D Design' refers to the process of creating computer-generated 3-dimensional shapes or objects. These can be imaginary or real-life objects, characters or landscapes that are created using 3D modelling software. 3D Design is used to convey ideas and concepts, create products, enhance customer experience, impart knowledge, and so on.



Applications of 3D Design

- Product Design and Development
- Architecture and Interior Design
- Industrial Manufacturing
- Medicine
- Education
- Entertainment
- Film Animation
- Gaming

3D Modelling

'3D modelling' refers to the process of creating a three-dimensional model of an already existing product. It is used to modify an existing product, develop a virtual representation of a product, or create a prototype of a product.

The Architecture, Engineering, and Construction (AEC) industry is the biggest market for 3D modelling, as it helps to improve design accuracy and reduce errors in the designs. This facilitates better and more effective communication of the design intent. It also helps in making more informed decisions throughout the construction process. 3D modelling helps engineers or architects visualize the product, identify potential clashes, and modify details at an early stage.

In the medical industry, detailed models of organs are used for various medical purposes, such as planning medical surgery, prosthetics and implants, medical education, etc. Also, in other educational streams, 3D modelling is used to explain complex topics to students for better understanding.

Similarly, 3D modelling is used in the gaming and movie industries for creating realistic 3D models, environments, objects, models, or other props. It provides a realistic look to objects or environments, thus creating an immersive experience for the viewers.



There are various online marketplaces for 3D content, such as TurboSquid, CGStudio, SketchFab, etc. that allow artists to sell their 3D models. Designing and modelling companies sometimes buy readymade models from these marketplaces instead of paying an employee to create 3D models from scratch.

3D Design vs 3D Modelling

3D Designing and 3D Modelling are similar terms and are often used interchangeably. Both require software to create 3D objects or models. However, both processes are different from each other.

3D Design	3D Modelling
'3D Design' is the process of creating a rough sketch or idea of a model from scratch to finish. It is the conceptualization or planning process of a new product.	'3D Modelling' is the process of creating a model of an already existing real-life object.
It is used for creating a prototype of a new product.	It is used to create a virtual model for modifying the existing products.

Designing and modelling artists conceptualise a product through 3D designing. This stage involves the creation of the layout, shape, and size of the product. Once the visual of the product is finalised, 3D modelling begins. 3D modelling is done using a software. It helps in applying texture, animation, lights, etc. A well-designed 3D model can enhance the overall aesthetic of the project.

Imagine you want to create a fun character for a video game. The character can be initially sketched on paper, but a 3D model must be made of it if you want to make it come to life in the game. To design the 3D model, you can use specialised software. You can start with the basic shapes of the character and add features like arms, legs, and a head. To make the character appear more lifelike, you can apply colours and textures.

Once you have created the 3D model of your character, you can then use it in your video game. Without 3D modelling, it is difficult to create realistic characters and environments in video games and other digital media.

We will use Blender to understand and explore 3D modelling features that will enable us to create more detailed 3D designs and models.

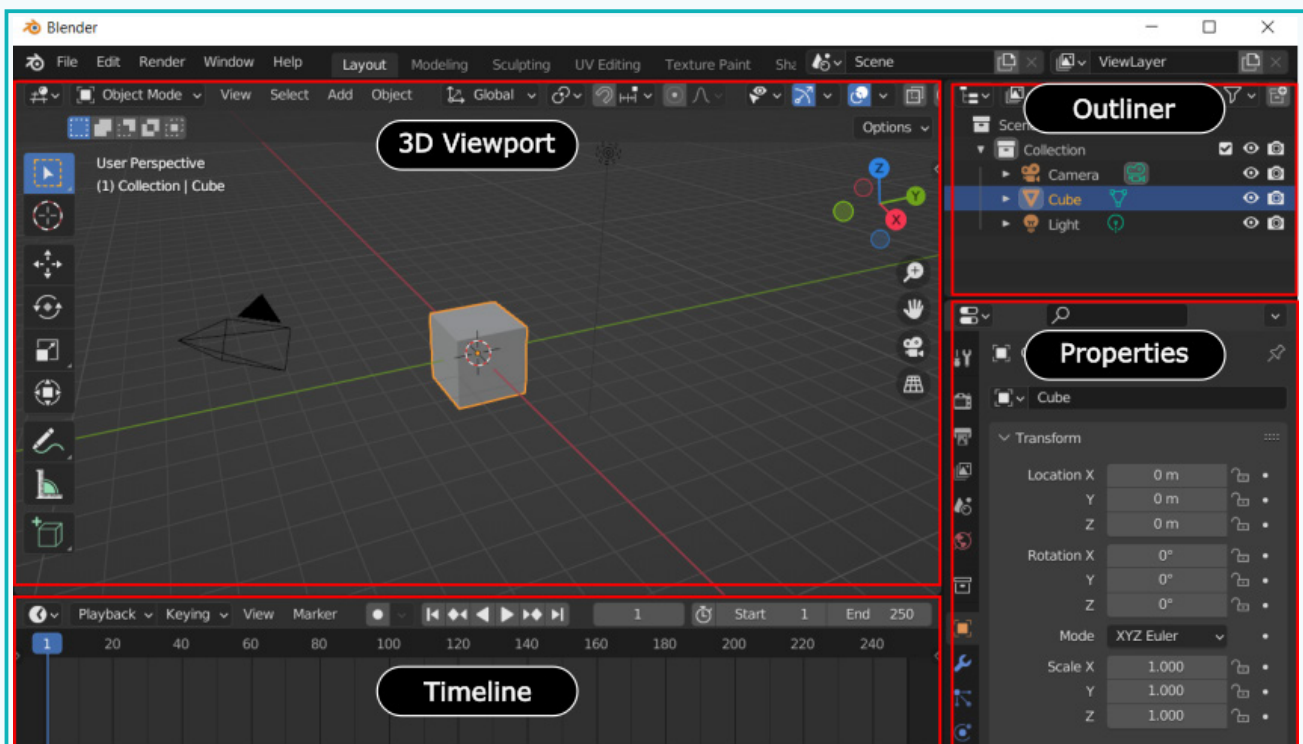
Activity

Explore the Workspace

First, let us explore the various components of the Blender Interface.

The Blender interface consists of several windows. Blender's default start-up layout consists of the following Editors:

- 3D Viewport (top left)
- Properties (bottom right)
- Outliner (top right)
- Timeline (bottom left)



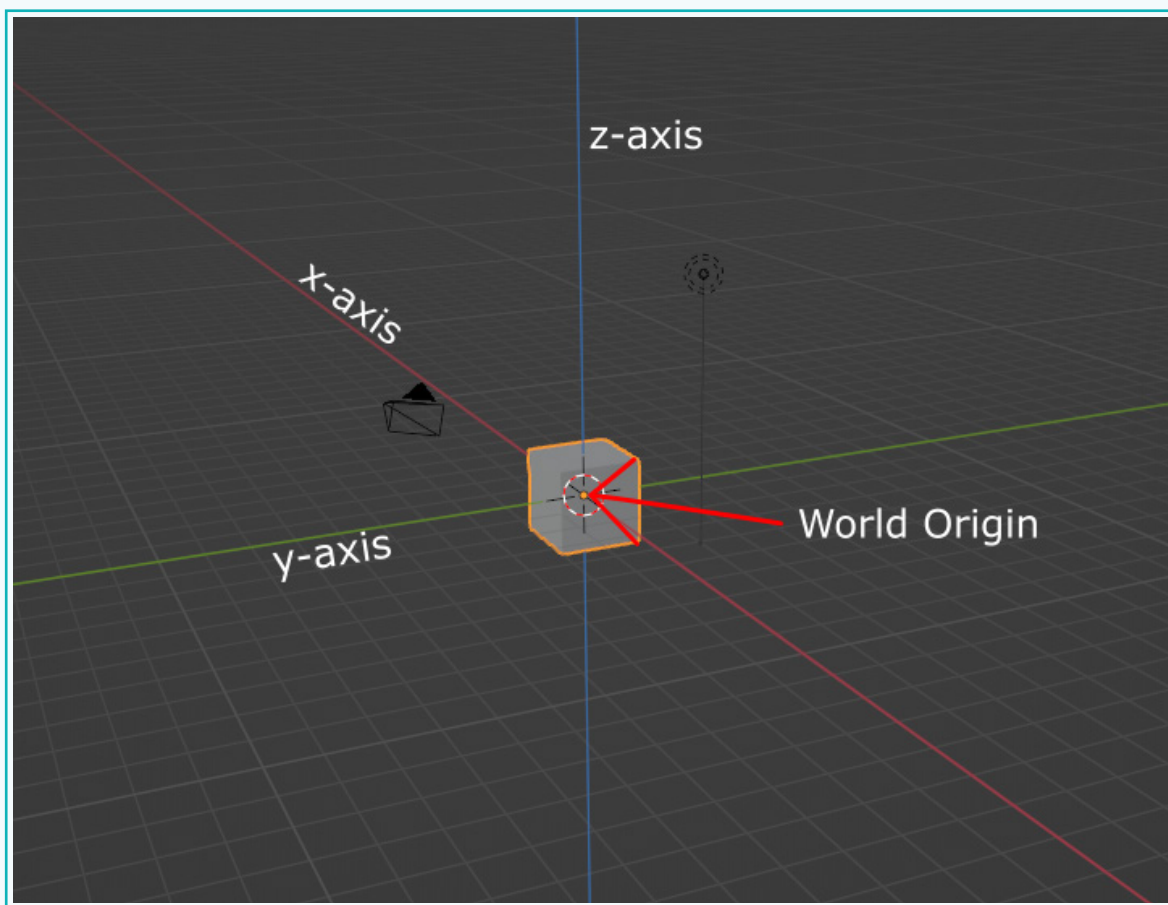
The **3D viewport** is the main window where 3D designs are created, edited, and animated. It is used to interact with the 3D scene for a variety of purposes, including modelling, animating, and texture painting. The Outliner in Blender is a list that organizes objects present in a scene and can be used to view, select and deselect, hide or show the objects.

Activity

The **Properties** window contains information about the selected object and some general scene properties. It has several tabs, each with a specific group of parameters, that allow you to edit active data.

The **Timeline** window is a fundamental part of Blender's animation system. It allows users to view, edit, and navigate animation frames.

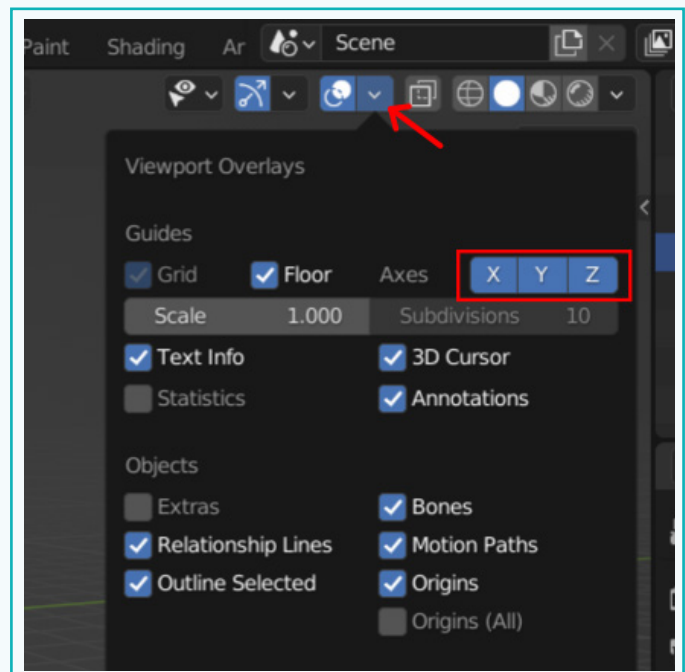
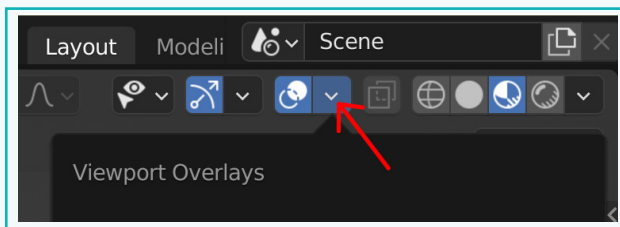
The **Viewport** also displays a grid floor. The lines forming the grid are composed of red, green, and blue lines that represent the world coordinate axes X, Y, and Z respectively. These 3 axes can define every possible point in three-dimensional space. The point where these 3 lines or axes intersect is the centre of the 3D world. This point is referred to as the 'World Origin' within Blender.



Activity

The visibility of the axes in the 3D viewport can be controlled through the **Overlays** menu. Clicking on the Overlays button will toggle the visibility of all overlay elements present in the scene.

Clicking on the arrow next to the **Show Overlays** button will display the Overlays menu. The visibility of individual overlay items in the 3D viewport can be controlled from this menu.



Being a 3D application, Blender allows us to view an object from all angles. Let us take a look at the gizmos which help us in achieving this.

What is a Gizmo?

In the literal sense, gizmo refers to a device or small machine which is designed to perform a particular task.

In Blender, a gizmo refers to a visual tool that helps users to manipulate objects in the 3D viewport. It helps us to perform transformations such as moving, rotating and scaling.

Some commonly used gizmos are the Navigation Gizmo and the Object Gizmo.

Activity

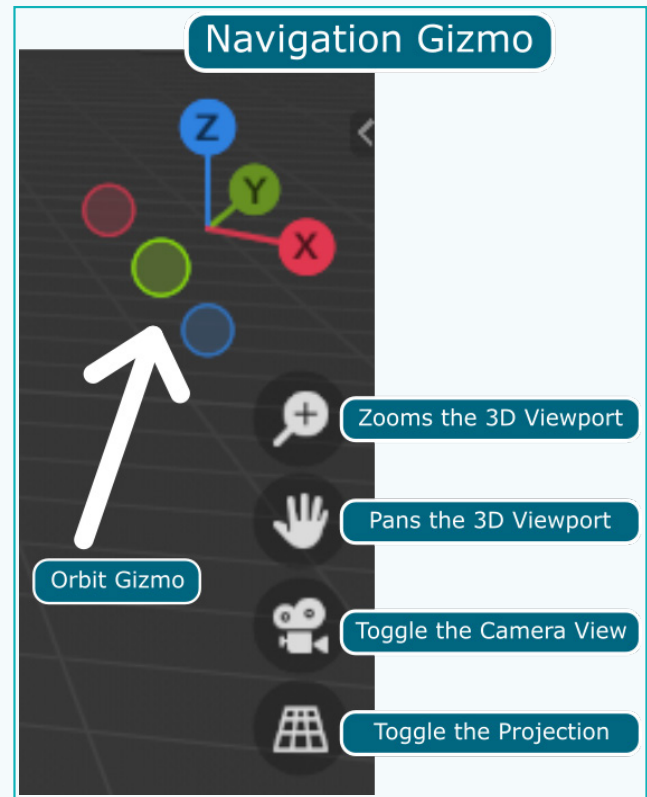
Navigation Gizmo

On the top right of the Editor, you can see the Navigation Gizmo. It consists of:

- Orbit gizmo
- 4 buttons below the Orbit gizmo.

The orbit gizmo shows the current orientation of the view. Dragging it with the Left Mouse Button orbits the view. Clicking on any of the axis aligns the view to the selected axis and clicking it again switches to the opposite side of the same axis.

The 4 buttons help to zoom, pan, toggle camera view and toggle projection.



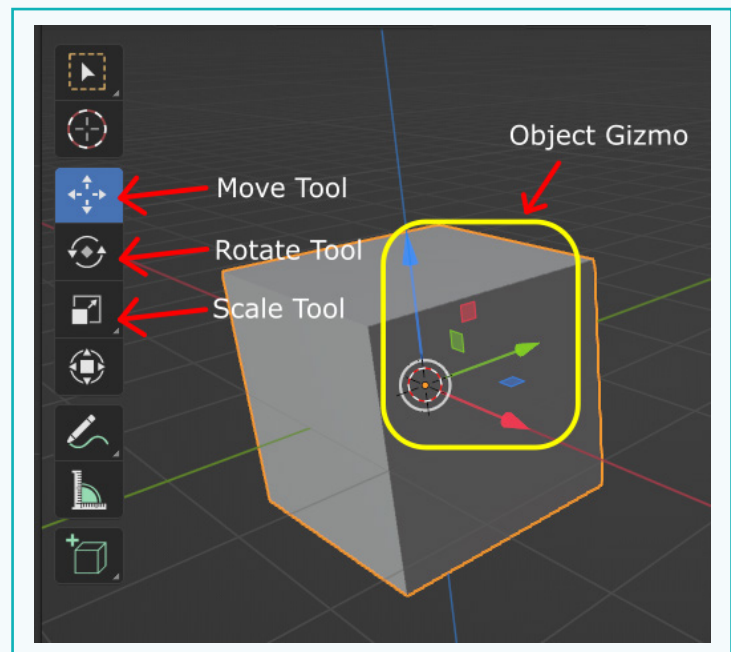
These can also be achieved through keyboard shortcuts, as shown below.

Function	Keyboard shortcut
Orbit	Click and Drag MMB
Pan	Shift + MMB
Zoom	Ctrl + MMB OR +/-

Activity

Object or Transformation Gizmo

This Gizmo is visible when any one of the Move, Rotate, or Scale tools is selected along with an object in the Viewport. It lets the user perform move, scale or rotate operations on the objects using the mouse on any of the 3 axes.



Below are few keyboard shortcuts for the commonly used actions.

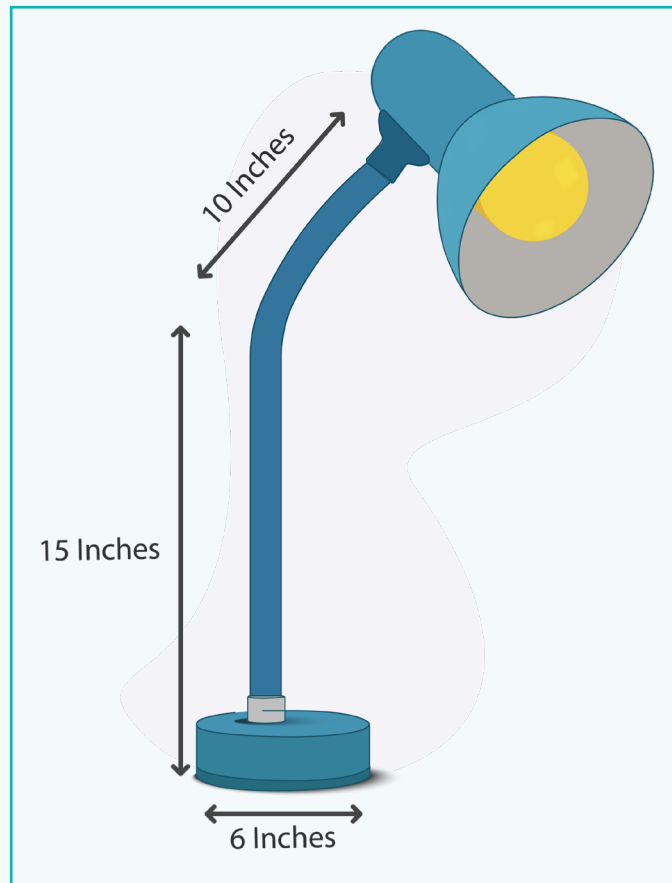
Function	Keyboard shortcut
Select	Left Click
Select All	A
Deselect All	Alt + A
Invert Selection	Ctrl + I
Hide Selection	H
Reveal Hidden Items	Alt + H
Toggle Left Sidebar Visibility	T
Reveal Right Sidebar Visibility	N
Frame Selected Object	.

Activity

We have already viewed the workspace where we will be working. Now, we will explore more modelling techniques by creating a real-world scale model of a table lamp.

Create the Base and Arm of a Table Lamp

We will use below image as a reference for creating our 3D model.

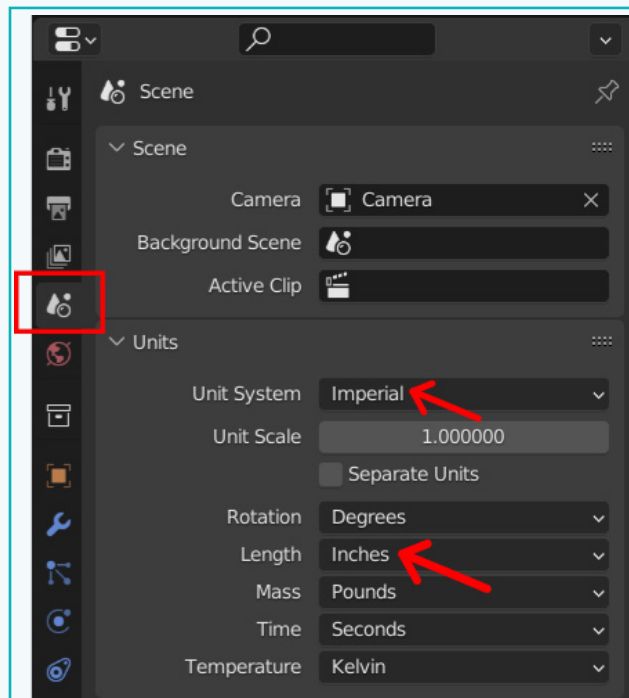


Note:

In Blender, the default unit of measurement is 'meters'. Let us change this to 'inches' so that we can easily specify real-life dimensions of a table lamp.

Activity

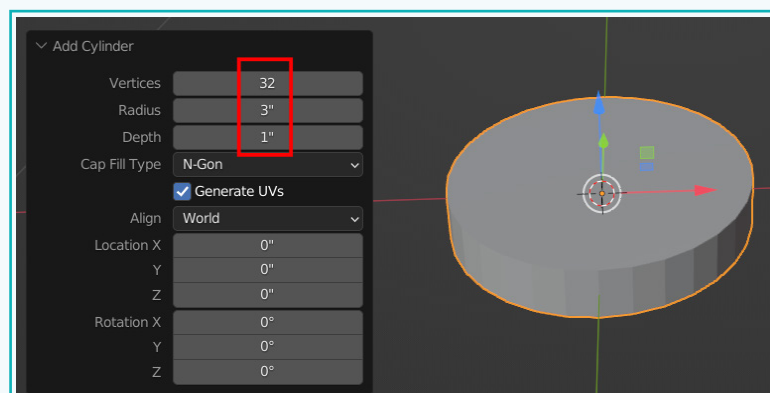
- 1 Click on the 'Scene properties' category in the 'Properties' window, and under 'Units', change the 'Unit System' to 'Imperial' and 'Length' to 'Inches'.



Let us begin with the base of the lamp.

In the reference image, the diameter is shown as '6 inches'. Let us start by creating a cylinder with a radius of '3 inches'. This will serve as the base for the lamp.

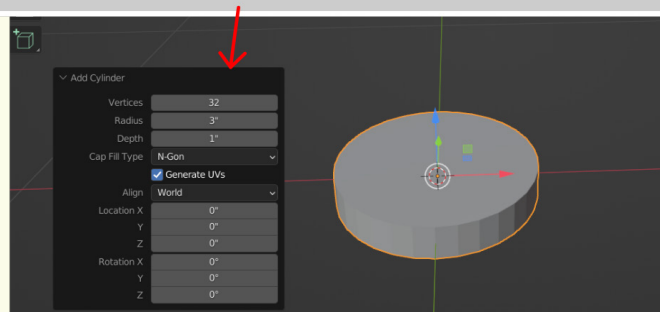
- 2 Create a 'cylinder' with the given dimensions.



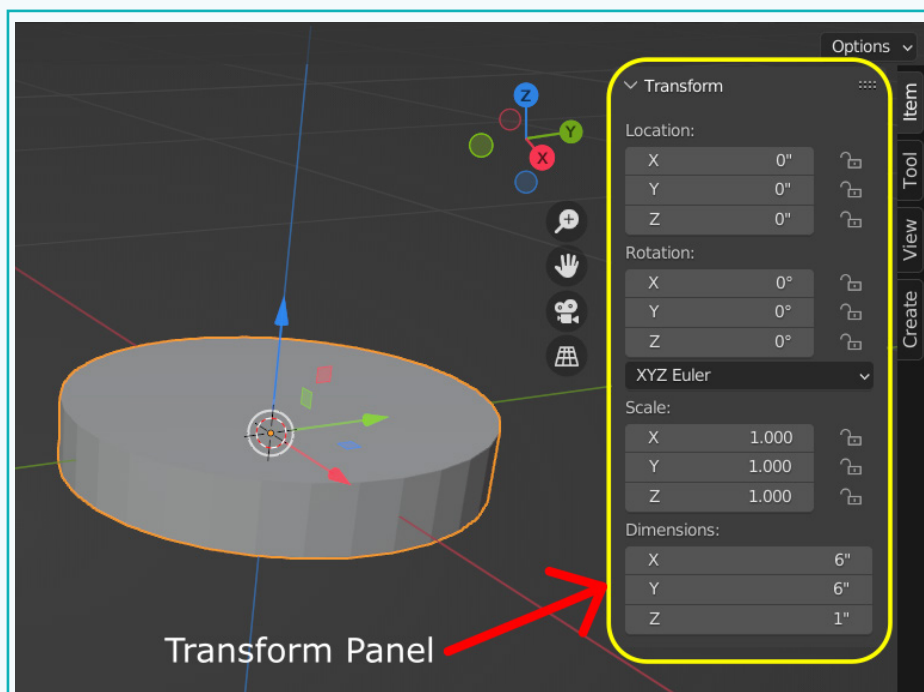
Activity

Note:

When you create a new object, the options menu appears at the bottom left of the 3D viewport. You can specify the dimensions and other properties in this menu. It appears immediately after the object is created. And goes away on clicking once anywhere in the viewport.

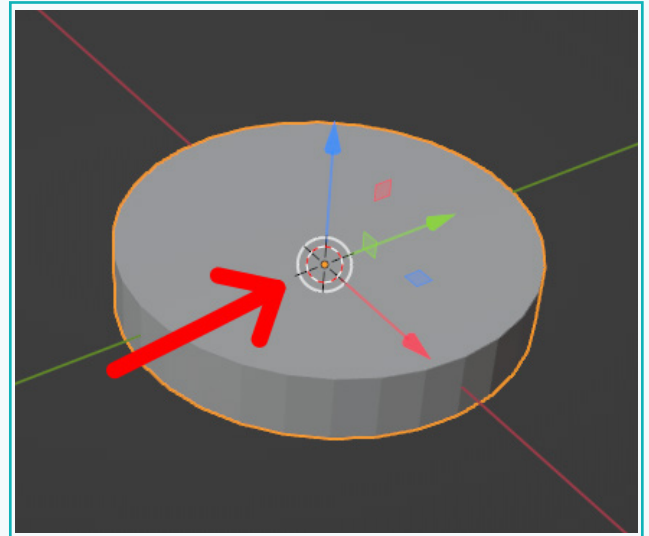


The dimensions and other transform details can be manipulated later using the 'Transform' panel. Press 'N' to view the 'Transform Panel' on the right of the 3D Viewport.



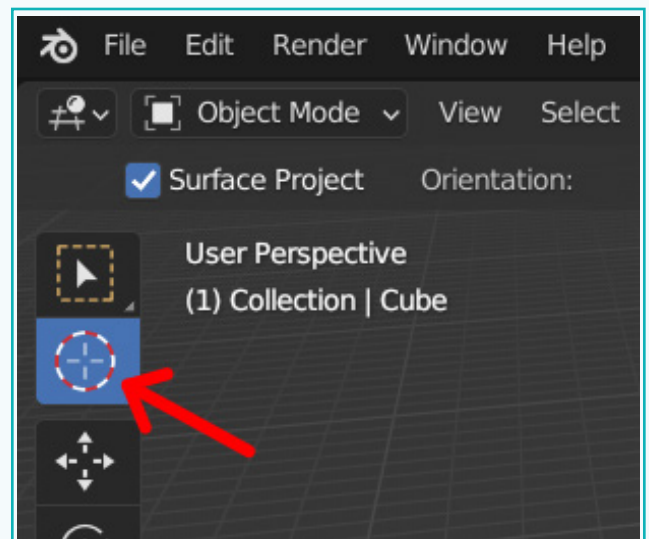
Activity

The cylinder gets created at the location of the 3D cursor. Any new object that we create will always be placed at the location of the 3D cursor.



3D Cursor:

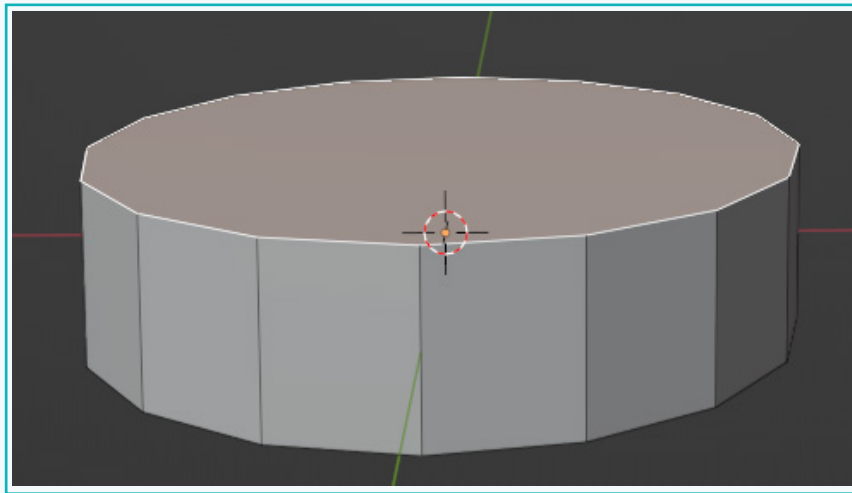
3D cursor is a multi-purpose tool that plays a vital role in the workflow within blender. By default, it is positioned at the centre of the world (World Origin). The 3D cursor's location can be changed by using the 3D cursor tool or by pressing 'Shift + RMB' (Right Mouse Button).



The 3D cursor can be positioned at very specific selected locations. This is crucial when we need to change the pivot point of objects. We will explore this in more detail later.

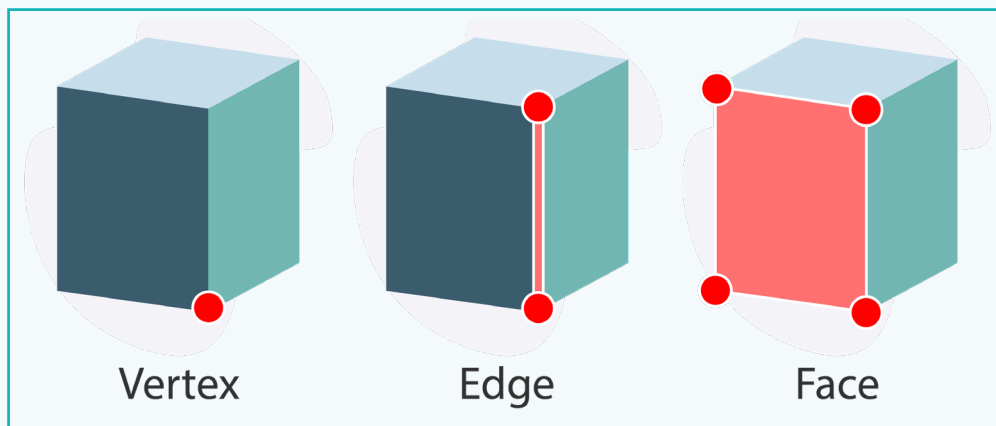
Activity

- 3 Press 'Tab' to go into the 'Edit' mode. Press '3' for the 'Face Selection' mode and select the top face, as shown.



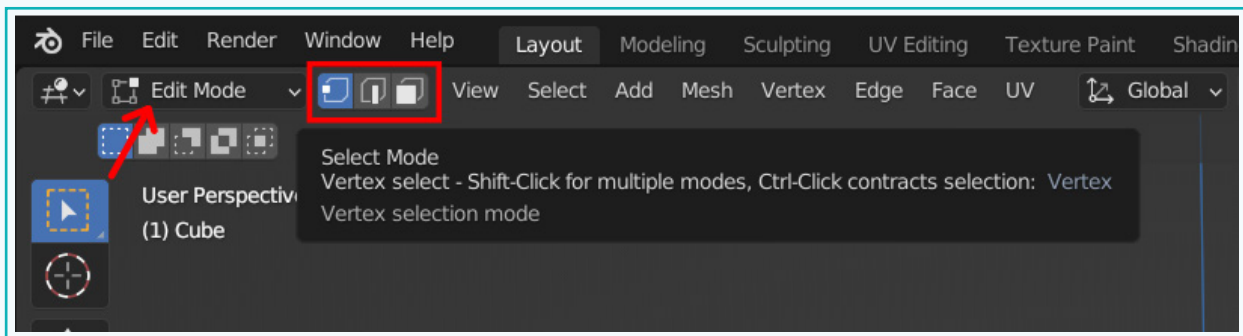
Recap

Every 3D model is made up of three elements – vertices, edges, and faces. A **vertex** is a point in space. When you connect 2 vertices, you get an **edge**. When you connect 3 or more vertices in a closed loop, you get a **face**.



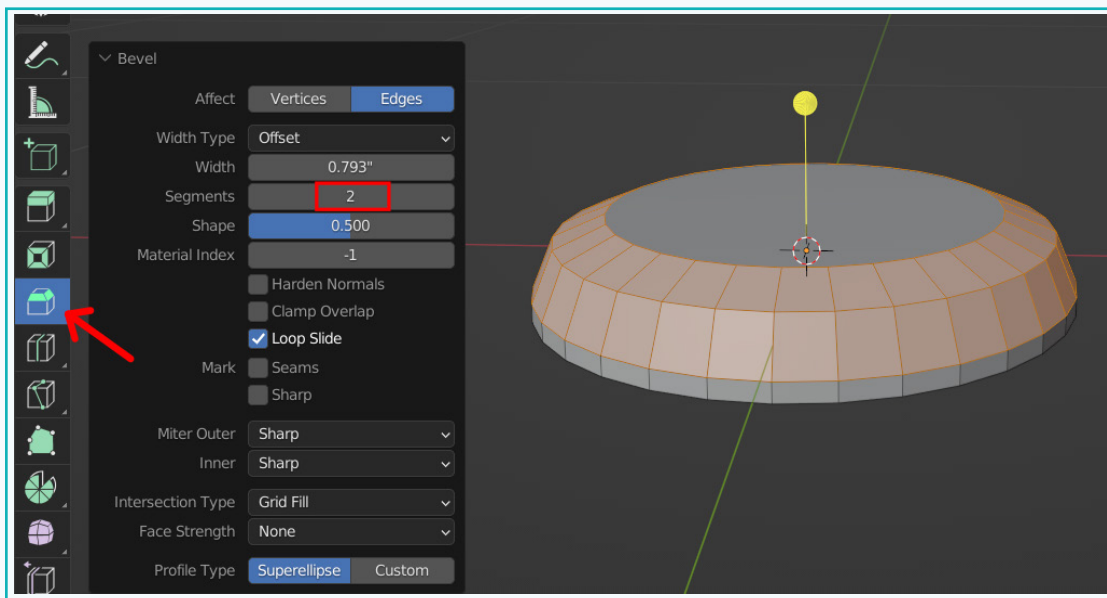
Activity

You can individually select vertex or edge or face selection by going into the 'Edit' mode and using the 'Select Mode' tool located at the top left of the viewport.



Alternately, you can use keyboard shortcuts 1, 2, and 3 for vertex, edge, and face selection, respectively.

- 4 With the face selected, click on the 'Bevel' tool located in the Toolbar on the left. The keyboard shortcut for this is 'Ctrl + B'.

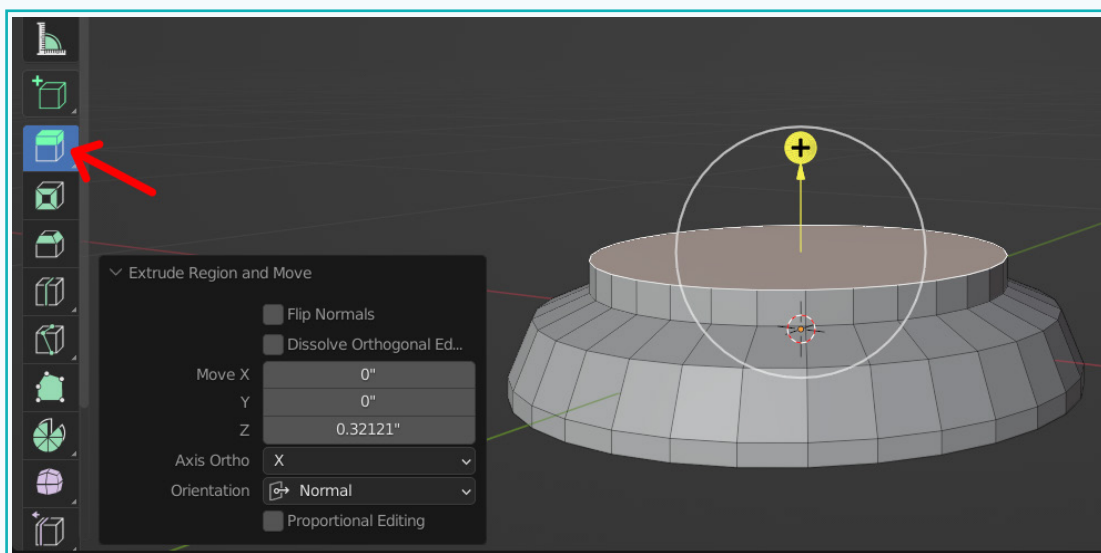
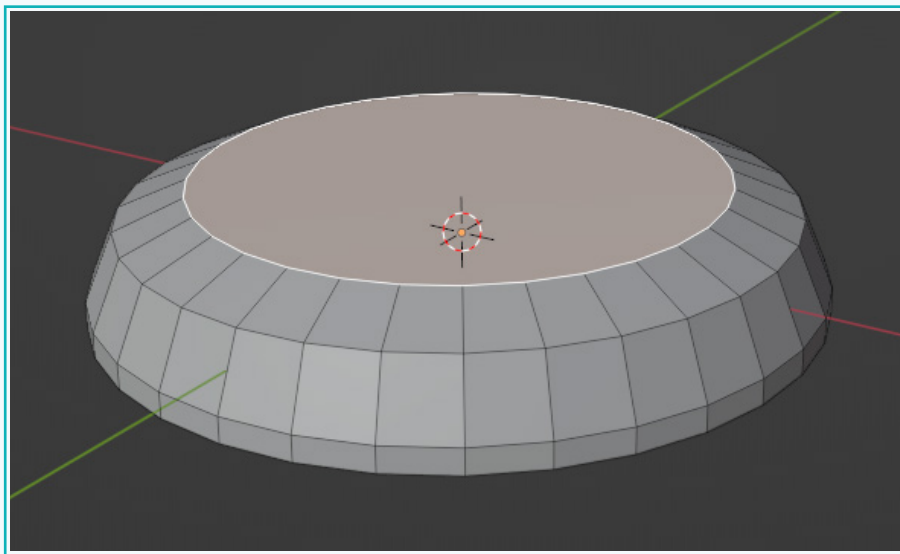


As we can see, the 'Bevel' tool creates rounded corners on the selected geometry. It smoothens out edges and corners. A higher 'Segments' value in the 'options', makes the geometry smoother.

Activity

- 5 Next, select the top face again. Select the 'Extrude' tool in the Toolbar to extrude the selected face upwards.

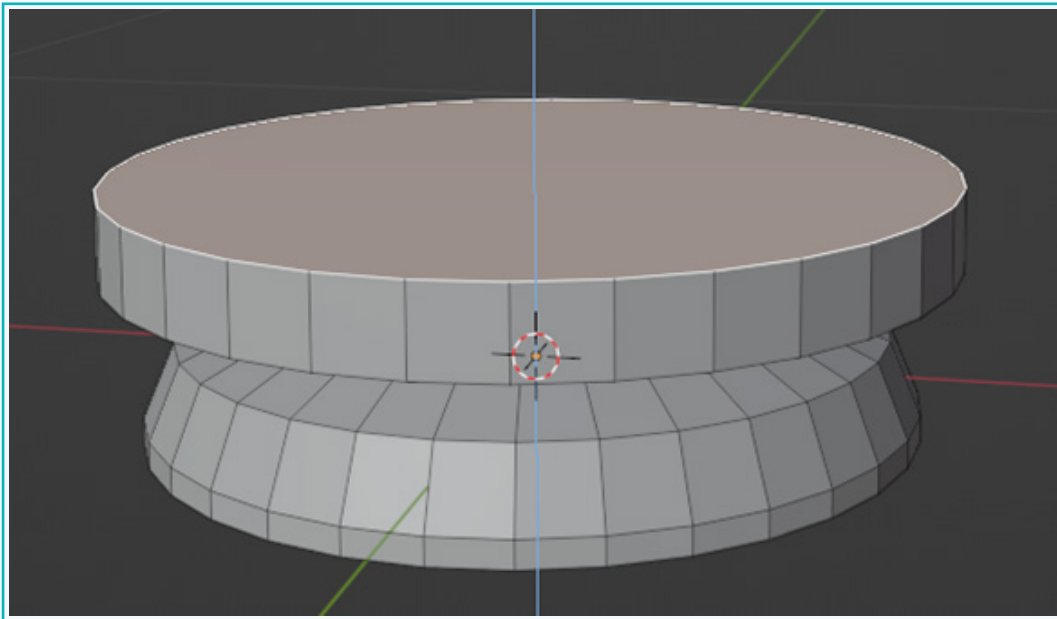
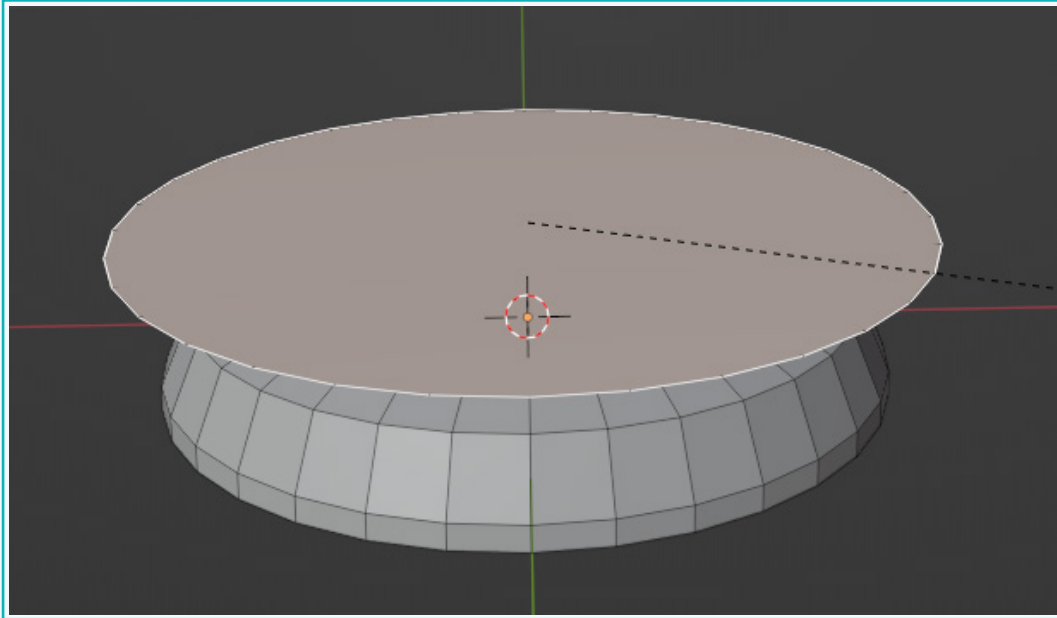
 - The keyboard shortcut for this is 'E', followed by X, Y, or Z, depending on which axis the extrusion needs to take place in.



The 'Extrude tool' is an 'Edit mode' tool meant to extend a selected portion of an object.

Activity

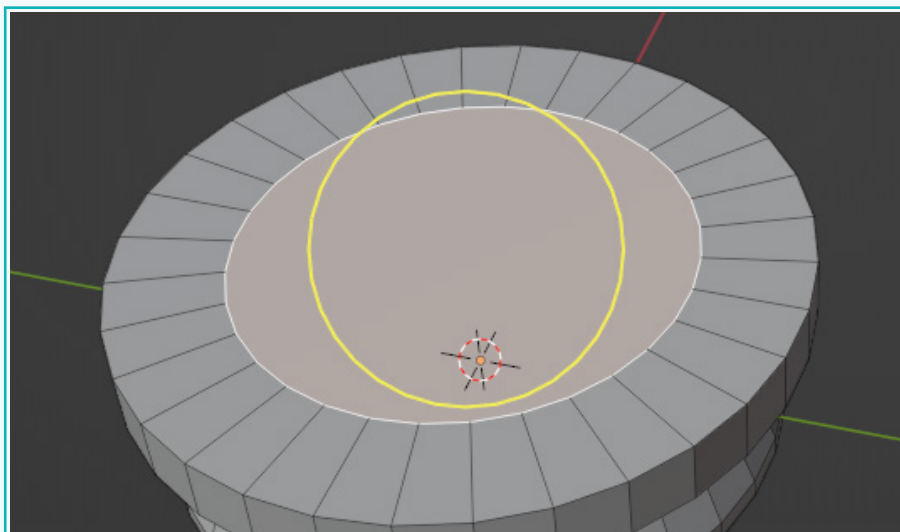
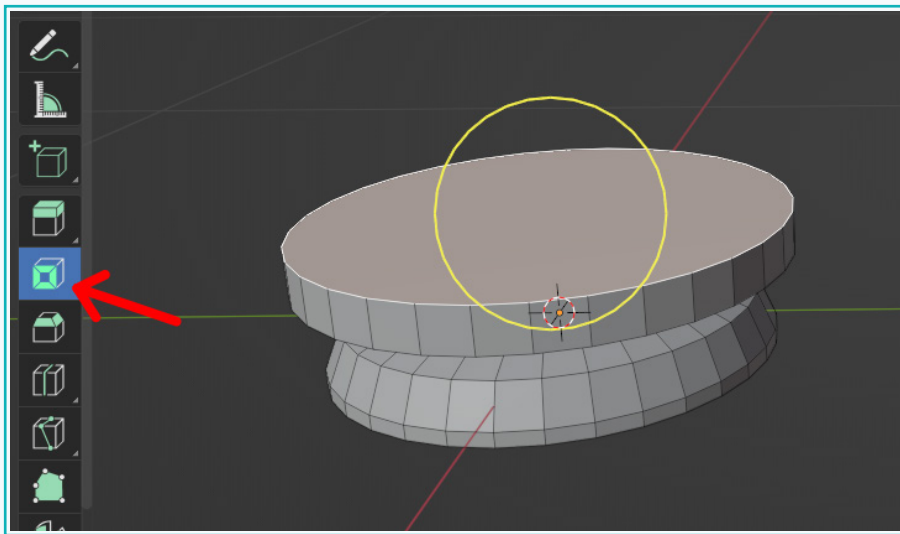
- 6 Press 'S' to scale the extruded face and press 'E' to again extrude the scaled face upwards.



Activity

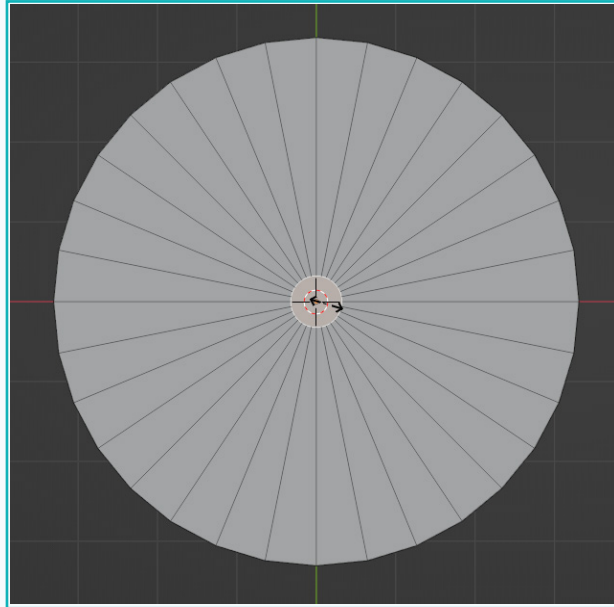
- 7 With the top face still selected, select the 'Inset' tool on the Toolbar. Create an inset of the selected face.

The 'Inset' tool is similar to the 'Extrude' tool, but the new face created is on the surface of the selection and does not change the surface's shape.

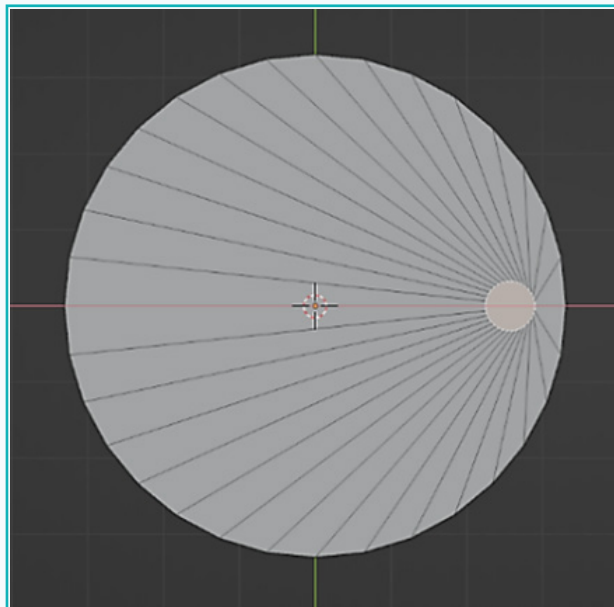


Activity

- 8 Scale down the inserted face, as shown, by pressing 'S'.



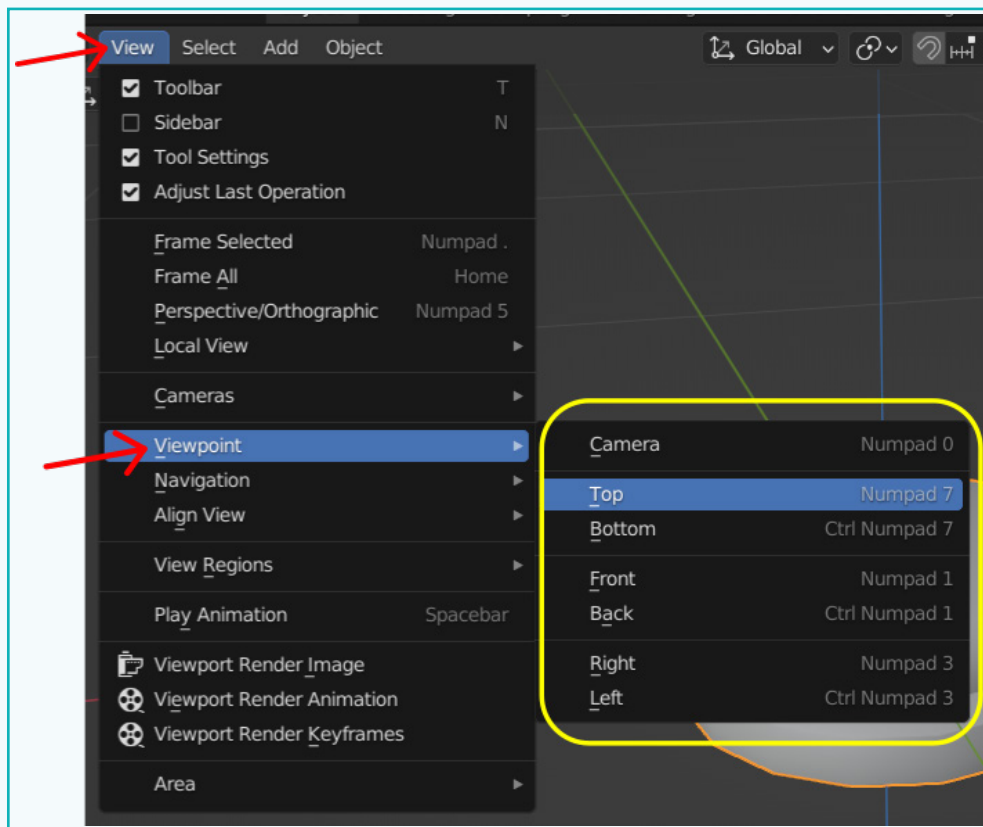
- 9 Go into the 'Top' view by pressing '7' on the numpad of the keyboard. Then, press 'G', followed by 'X' to move the selected face towards the right on the x-axis.



Activity

ViewPoints

'Viewpoint' under the 'View' option helps to align objects to specific views. The 'Orbit Gizmo' can also be used for this purpose.



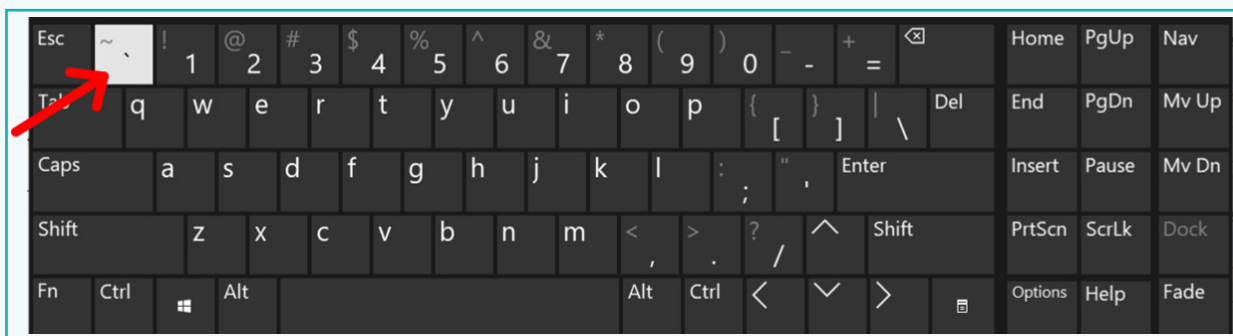
The 'Hotkeys' for the different views are as shown -

Top:	Numpad7
Front:	Numpad1
Right:	Numpad3
Bottom:	Ctrl- Numpad7
Back:	Ctrl- Numpad1
Left:	Ctrl- Numpad3

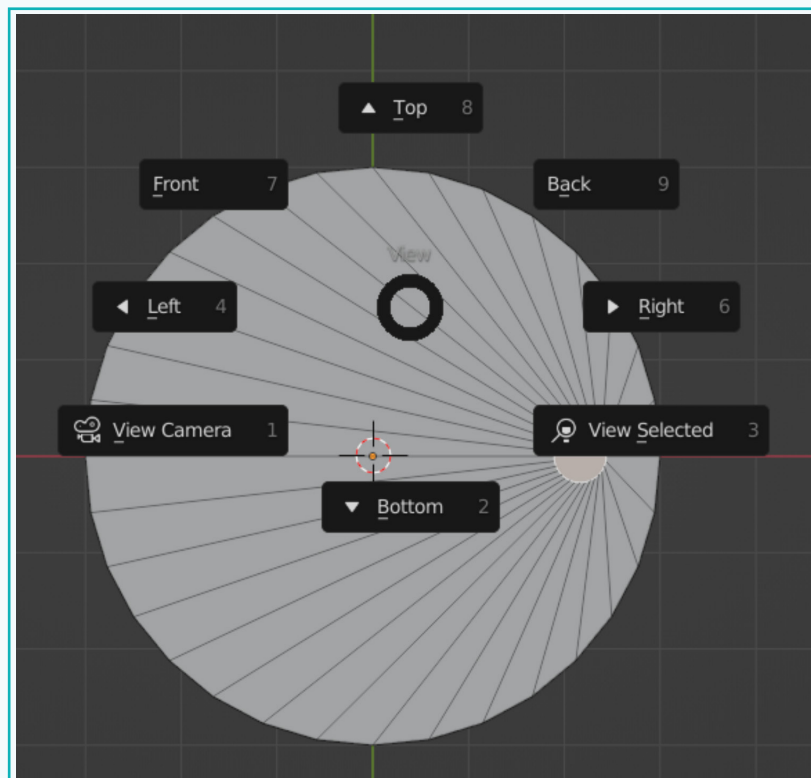
Activity

Note:

If the Numpad is not present, press the 'Backtick (`)' button on the keyboard.



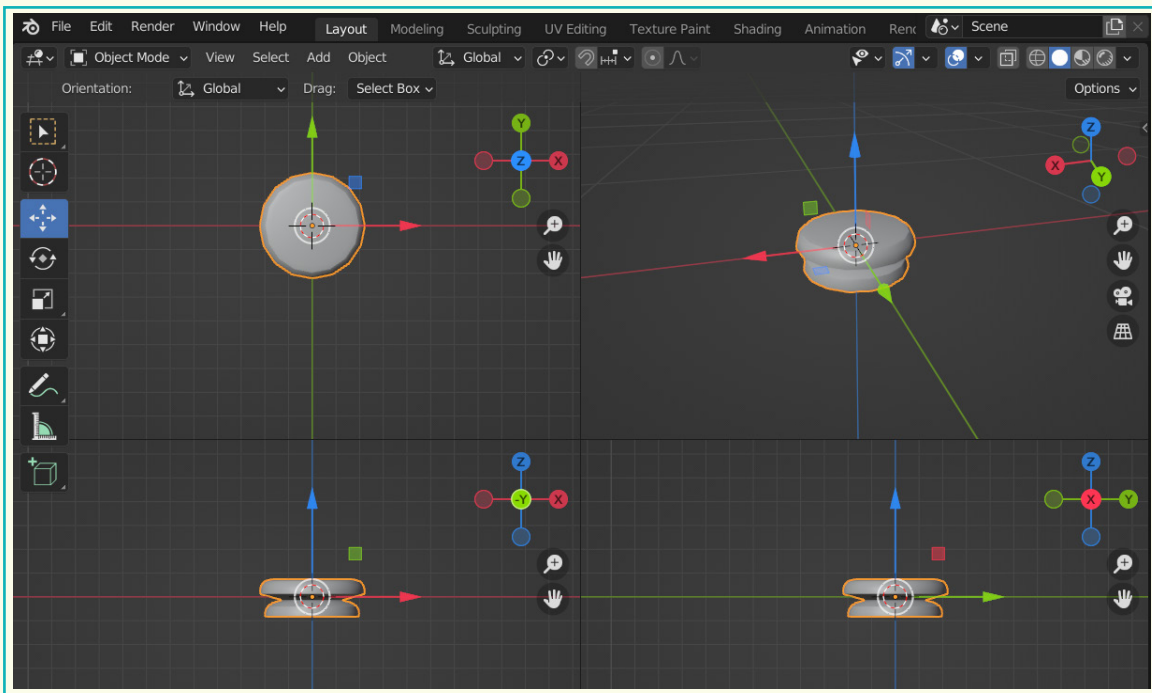
'Backtick' opens the 'View' pie menu which allows you to easily switch between different views.



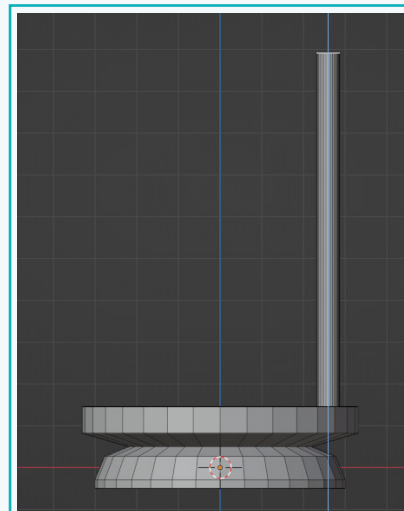
Activity

Note:

'Ctrl+Alt+Q' gives the **Quad View**. This view displays different views at the same time. This is very useful for making precise changes while considering different perspectives.

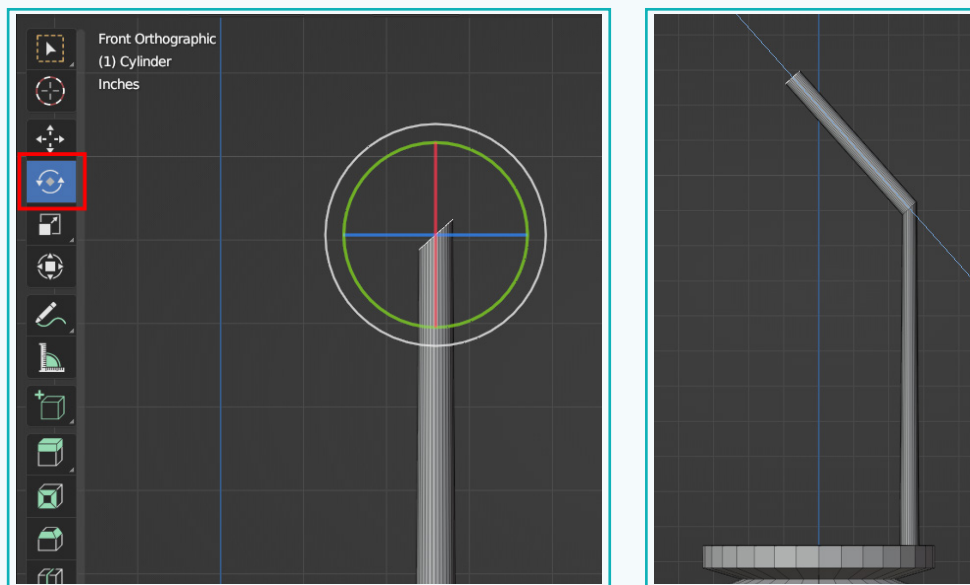


- 10 Go to the front view. Press 'E' to extrude the selected face upwards to create the neck of the lamp.

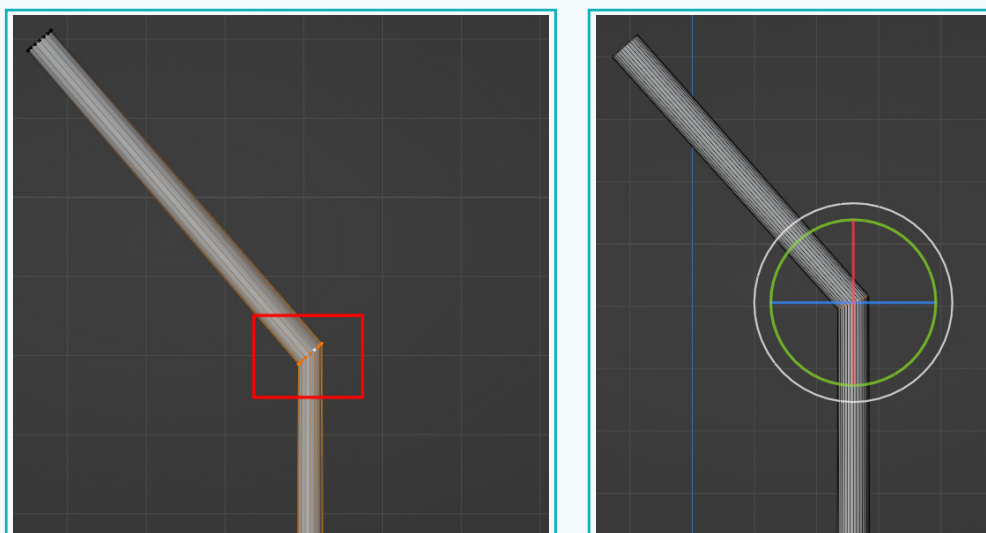


Activity

- 11 Press 'R' to rotate the face, as shown and then, press 'E' to extrude the face at an angle, as shown.

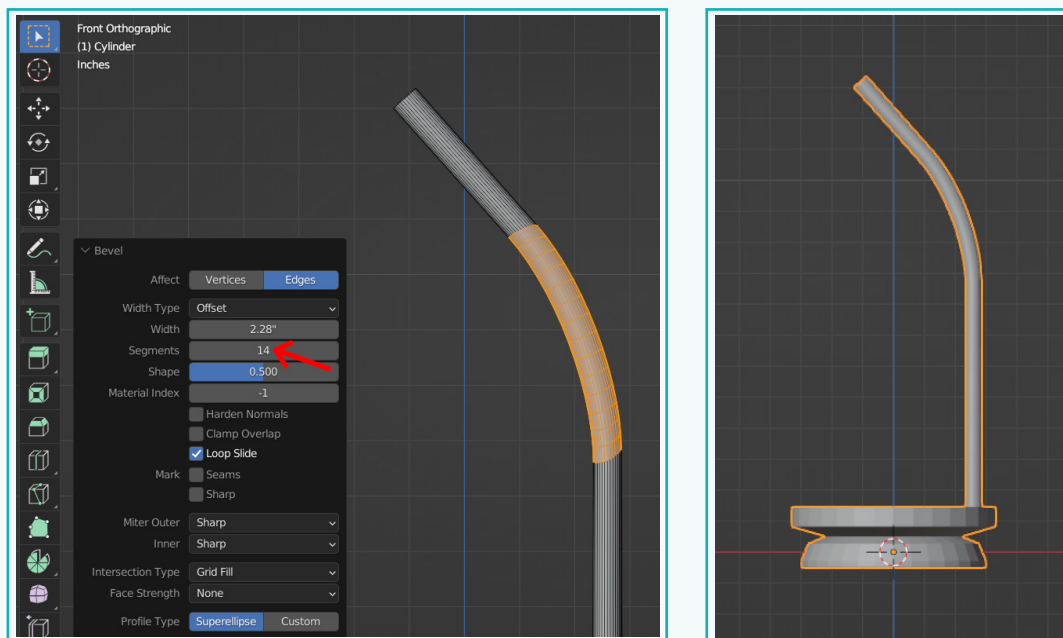


- 12 Select the 'Edge loop', as shown and rotate it slightly to ensure that the width of the cylinder is even at the bend.



Activity

- 13 Press 'Ctrl+B' to bevel the selected edge. Specify the desired number of segments to create a smooth bend for the arm of the lamp.



The base and lamp arm are ready.

Exercise

State if the statements are True or False

- 1** The 3D cursor's position is fixed and cannot be manipulated.
 _____
- 2** The 'Quad View' gives two different views at the same time.
 _____
- 3** The 'Viewpoint' menu as well as the 'Orbit Gizmo' can both be used to align objects with specific views.
 _____

Fill in the Blanks.

- 4** The _____ Gizmo helps to align objects with specific views.
- 5** The _____ tool helps to smooth out edges and corners.
- 6** The _____ tool helps to extend selected parts of geometry.
- 7** When you connect 2 vertices, you get an _____.
- 8** A 'vertex' is a _____ in space.

Exercise

 Answer the questions with one sentence

9 What is the 'Bevel' tool used for?



10 How is the 'Inset' tool different from the 'Extrude' tool?



 Tick mark ☒ the Correct Answer

11 The Orbit Gizmo -

- a. toggles between the 'Object' and 'Edit' mode. ☐
- b. shows the current display mode. ☐
- c. shows the current orientation of the view. ☐
- d. allows you to move, scale, and rotate objects using the mouse. ☐

12 Which option controls the visibility of the 3 axes in the 3D viewport?

- | | | | |
|---------------------|--------------------------|-----------------|--------------------------|
| a. Viewport Shading | ☐ | b. Gizmo | ☐ |
| c. Overlays | ☐ | d. Object Types | ☐ |

13 The settings for changing the unit of measurement are present in -

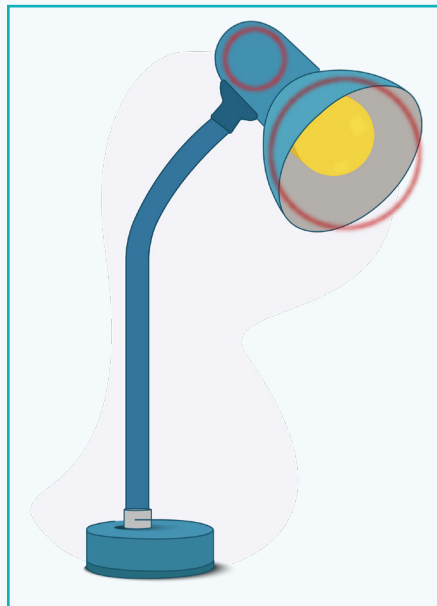
- | | | | |
|----------------------|--------------------------|------------------------|--------------------------|
| a. Scene Properties | ☐ | b. Material Properties | ☐ |
| c. Object Properties | ☐ | d. World Properties | ☐ |

Activity

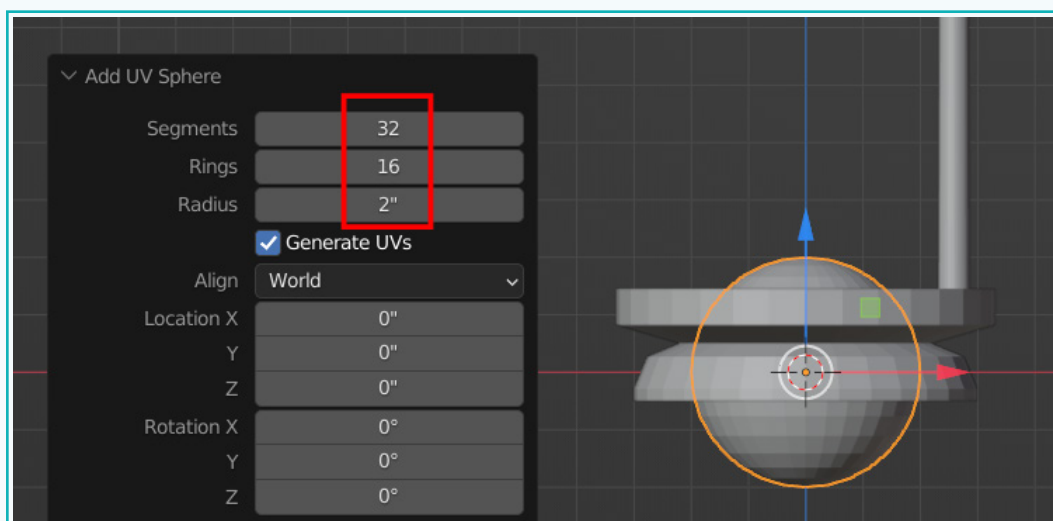
Create the Lampshade

Next, let us create the Lampshade.

The upper parts of the lampshade can be visualised as hemispheres (half sphere). Hence, let us try to model it using UV Spheres.



- 1 Create a UV Sphere with given dimensions.

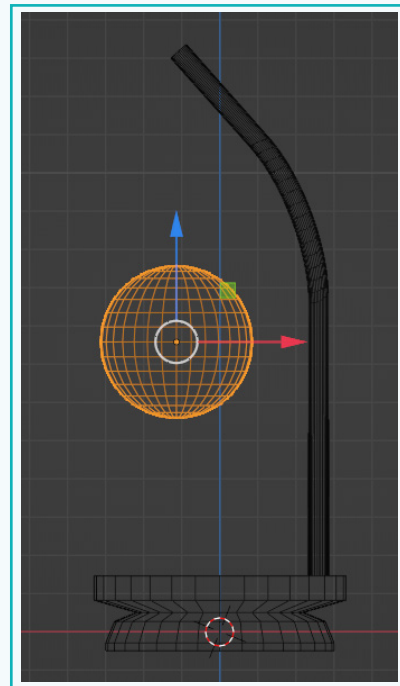
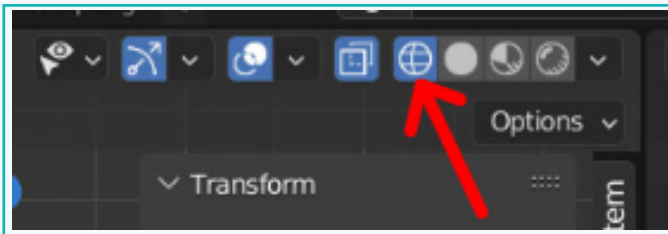


Activity

- 2 Press '1' for 'Front View' and go into the 'Wireframe Shading' mode.

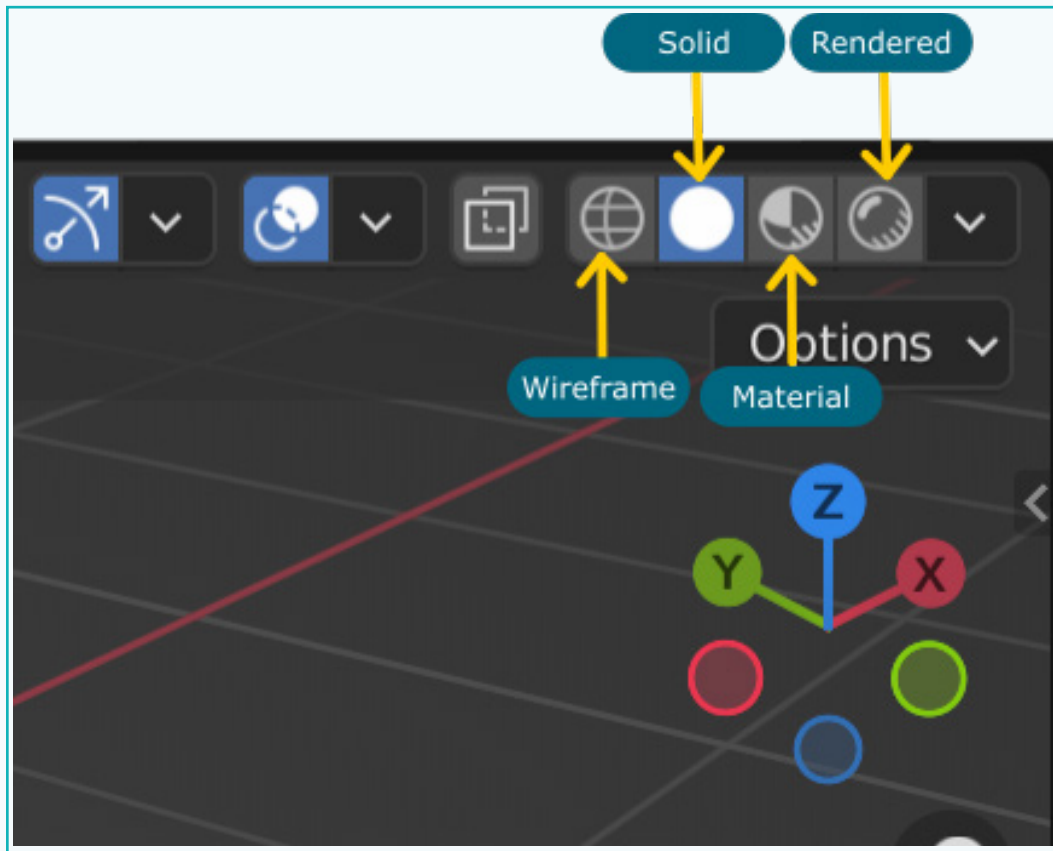
Note:

'Wireframe Shading' mode helps to get a good view of the geometry. It also helps to view and select even those vertices/edges/faces which are not visible from the current view.



Viewport Shading – 'Shading' refers to the way objects are displayed and lit in the 3D viewport. There are different shading modes available in Blender that help with different tasks.

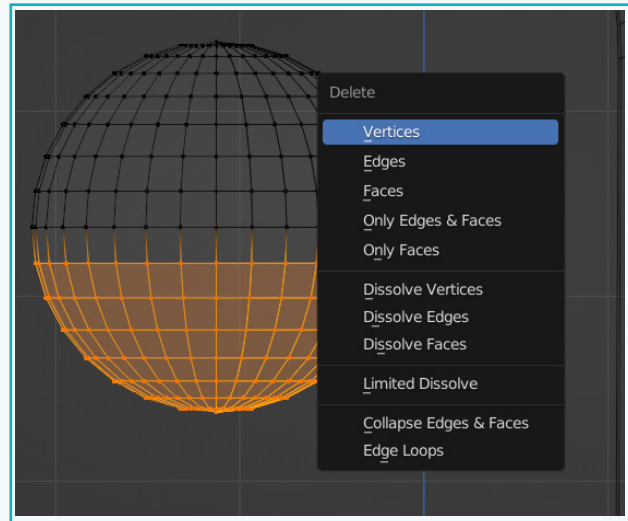
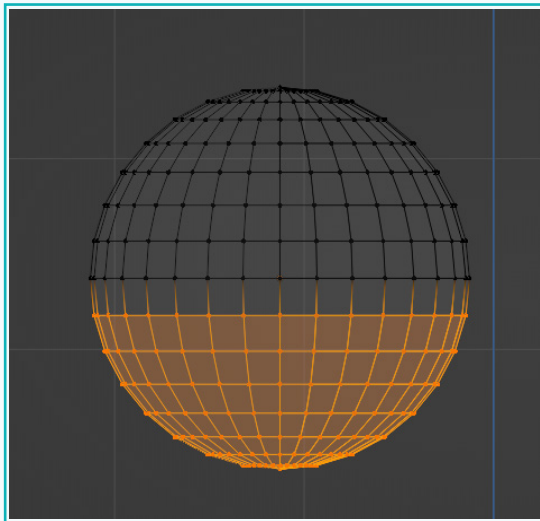
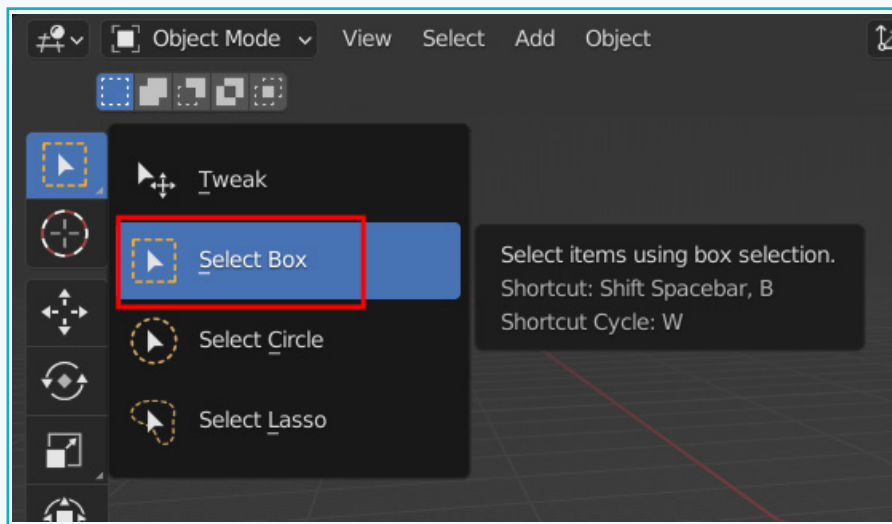
Activity



Wireframe	Only displays the edges of the objects in the scene
Solid	Displays solid geometry with simplified shading and lighting
Material	Useful for previewing materials
Rendered	Displays the preview of the final render, including scene lighting effects

Activity

- 3 Select the bottom half vertices/edges/faces using 'Select Box'. Delete the bottom half.

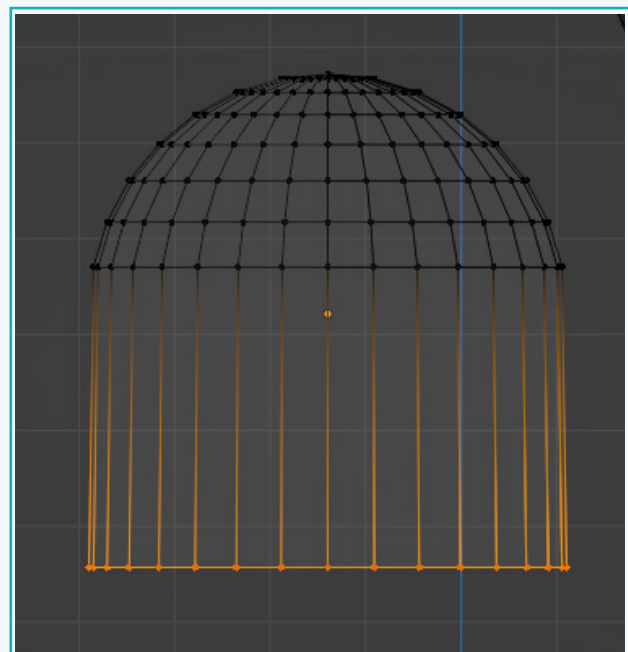
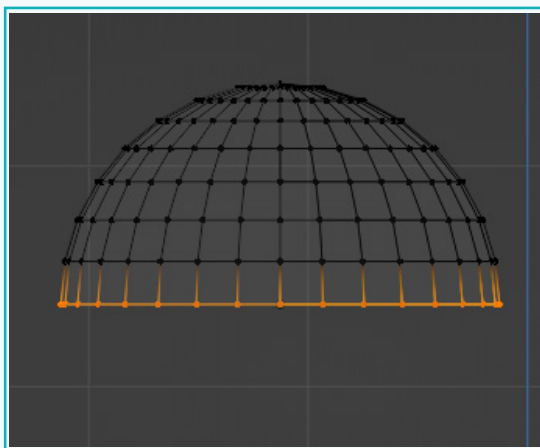


Activity

Different Selection Modes:

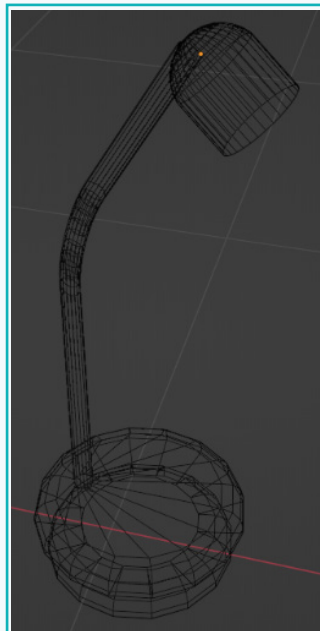
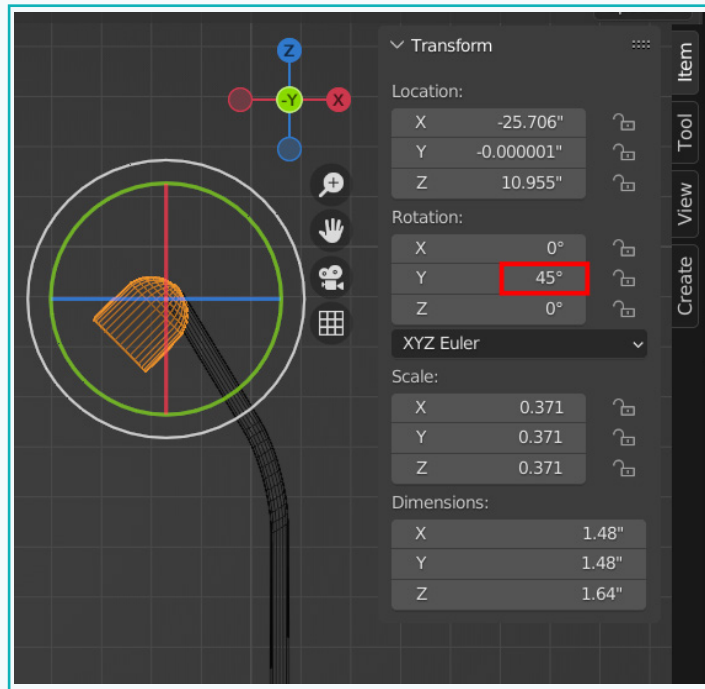
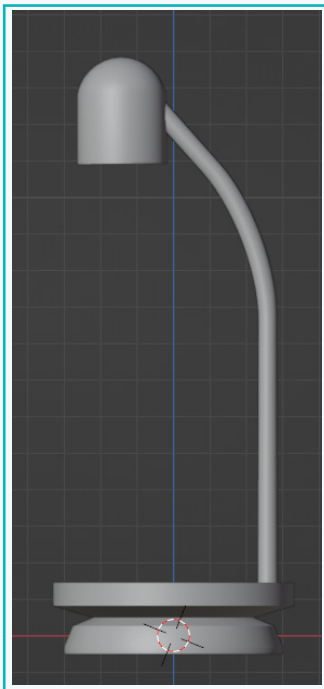
Tweak	Drag to select an item and drag it around.
Select Box	Drag to create a rectangle and select all the items within the rectangle.
Select Circle	Drag to create a circle and select all the items within the circle.
Select Lasso	Drag to create a freeform shape and select all the items within it.

- Select the bottom vertex loop by double-clicking on any one vertex in the loop. Move the selected vertex loop downwards to create the top part of the shade. This can be done using the 'Move' tool or by pressing 'G' followed by 'Z' to move the vertex loop downwards along the z-axis.



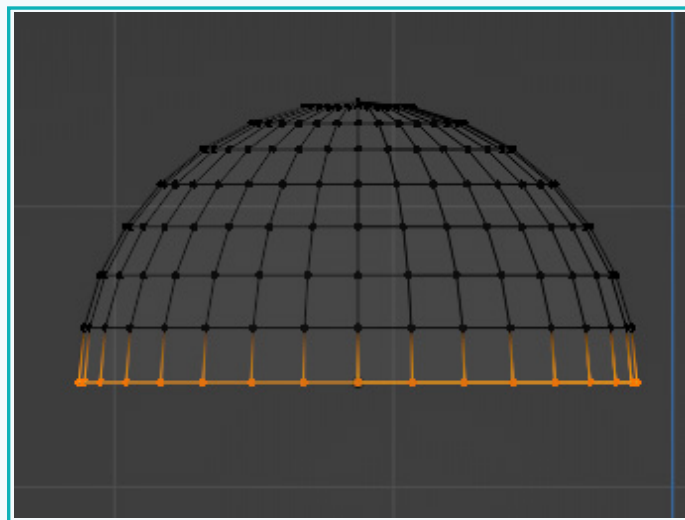
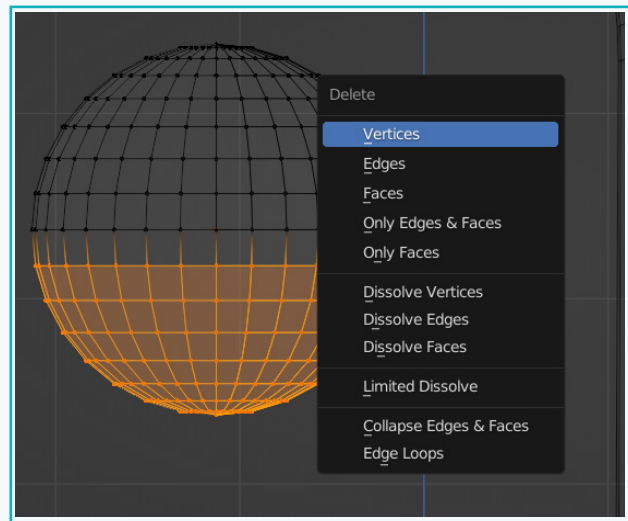
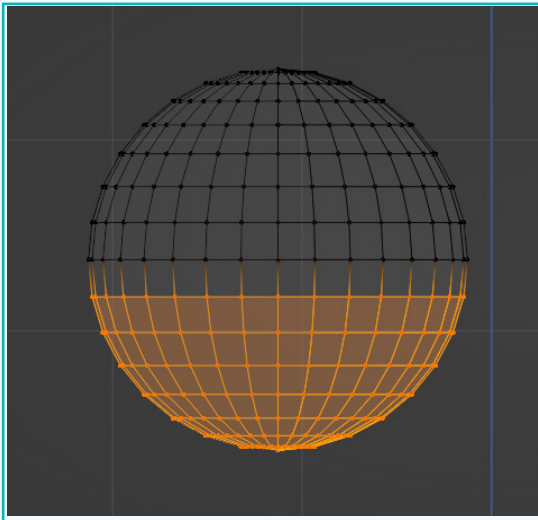
Activity

- 5 Go into the 'Object' mode and position, as shown. Rotate on the y-axis by '45°'. Go into the different orthographic views to check if the positioning is proper.



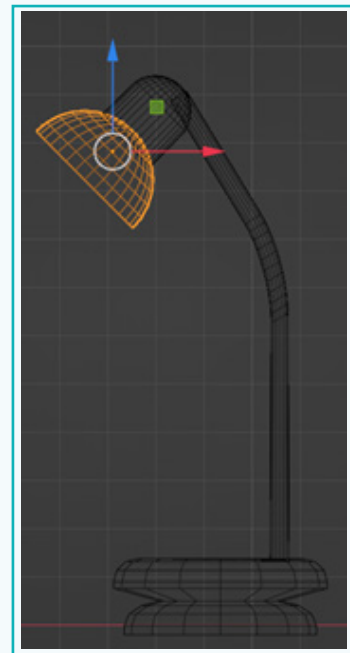
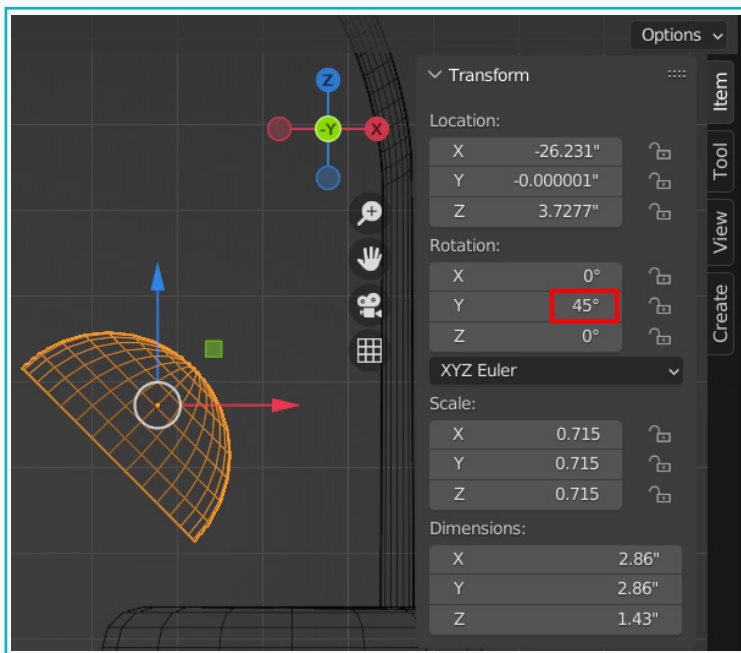
Activity

- 6 Create another UV Sphere, go into the front view and the wireframe mode, and delete the bottom half of the sphere, as done in the previous step. Use this as the remaining bottom part of the shade.

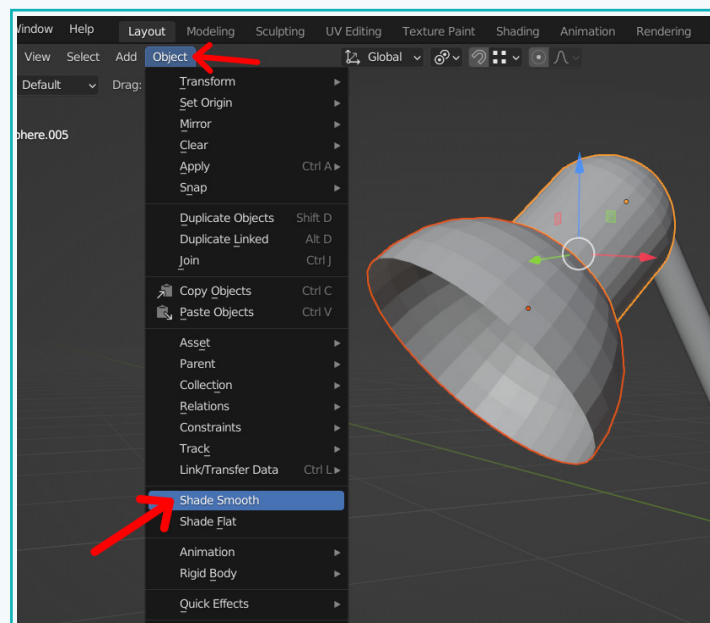


Activity

- Go into the 'Object' mode and rotate it on the y-axis by '45°'. Position appropriately to complete the model of the lamp shade.



- Select both the parts of the Lampshade and click on the 'Object' menu and select 'Shade Smooth'.

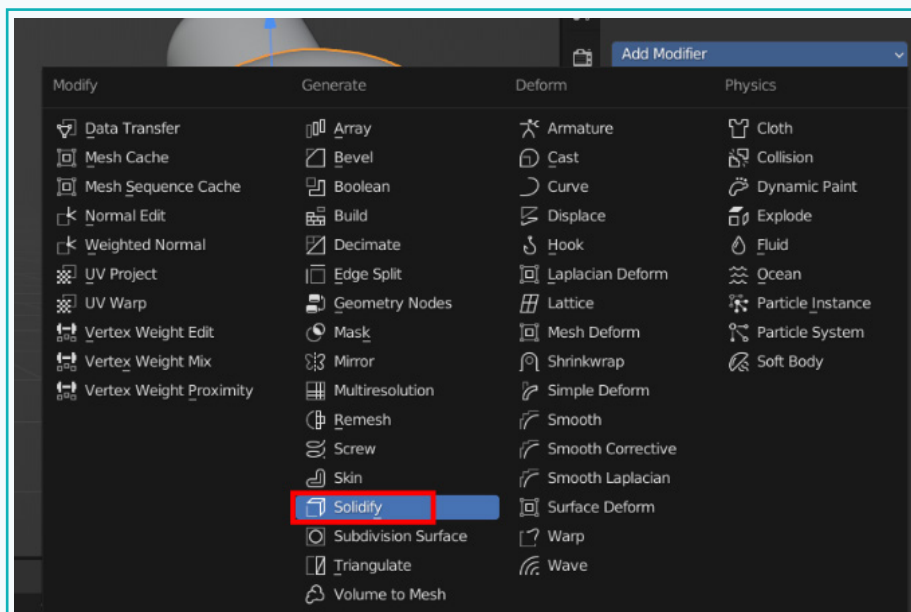
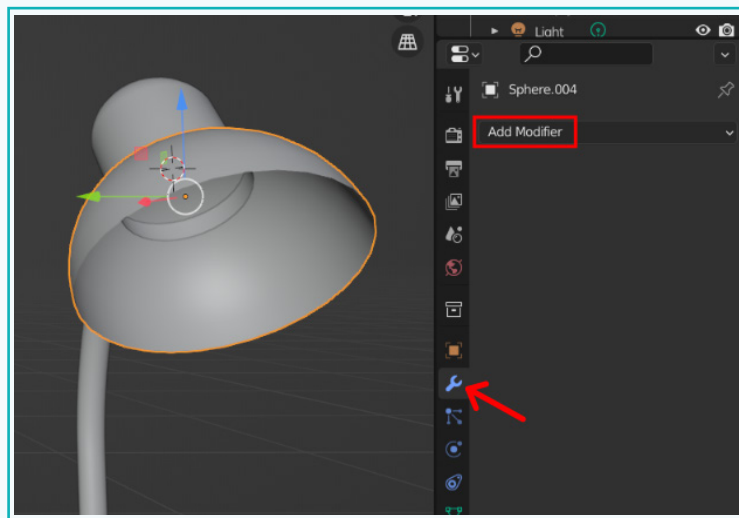


Activity

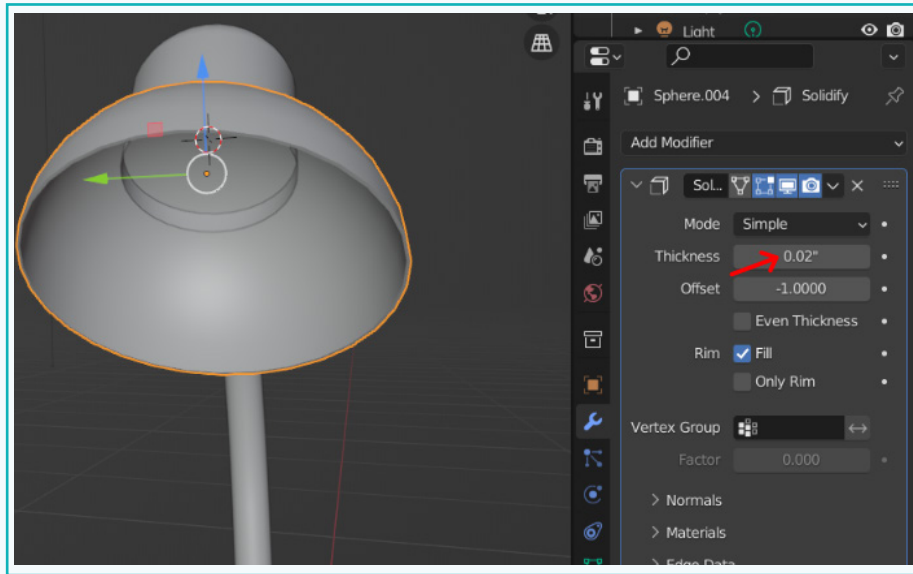
Note:

Multiple models can be selected by holding down the 'Shift' key and left clicking on the objects.

- 9 Go to the 'Modifiers' tab and add the 'Solidify' modifier. Change the 'thickness' property to add thickness to the model.



Activity



Output



Exercise

 Answer the questions with one sentence

14 Which mode is helpful in selecting parts of the geometry not visible in the current view?



15 Which modifier can be used to give thickness to a model?



 Tick mark ☒ the Correct Answer

16 Which key can be used to select multiple objects?

- | | | | |
|----------|--------------------------|----------|--------------------------|
| a. Ctrl | ☐ | b. Shift | ☐ |
| c. Space | ☐ | d. Alt | ☐ |

17 The _____ mode displays solid geometry with simplified lighting and shading.

- | | | | |
|--------------|--------------------------|-------------|--------------------------|
| a. Wireframe | ☐ | b. Solid | ☐ |
| c. Material | ☐ | d. Rendered | ☐ |

18 The _____ Selection mode can be used to drag a freeform shape and select all items within it.

- | | | | |
|------------------|--------------------------|-----------------|--------------------------|
| a. Select Box | ☐ | b. Select Lasso | ☐ |
| c. Select Circle | ☐ | d. Tweak | ☐ |

Exercise

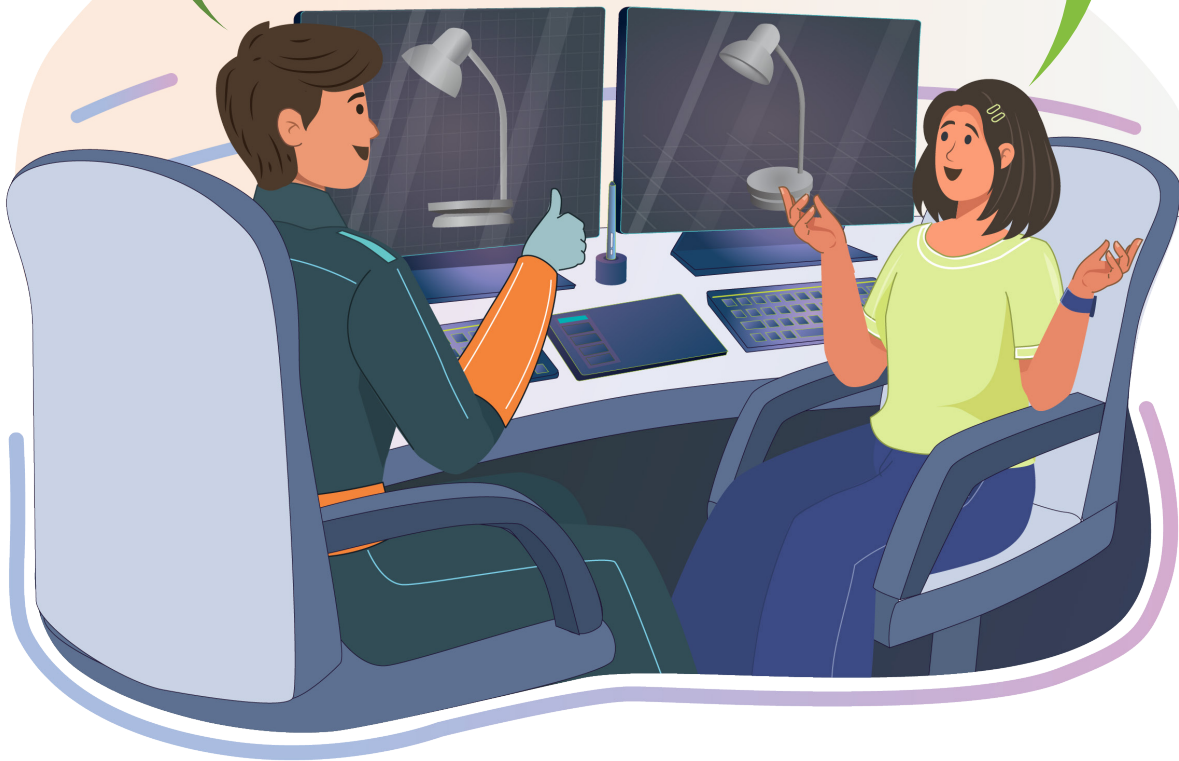
Find the Odd One Out

- | | | | | |
|-----------|---------------------|--------------------------|-----------------|--------------------------|
| 19 | a. Select Circle | ☐ | b. Select Box | ☐ |
| | c. Select Rectangle | ☐ | b. Select lasso | ☐ |
| 20 | a. Wireframe | ☐ | b. Solid | ☐ |
| | c. Material | ☐ | b. X-Ray | ☐ |

In this session, we have discussed 3D Designing and Modelling. We have learnt some new concepts of modelling in Blender and applied them to create the basic structure of a table lamp.

That's all for today, Tia! Be ready to explore some more interesting concepts in the next session.

Thank you DeZain! I can't wait to create more 3D magic and also finish my model.



Tech Flash

- **Orbit Gizmo** : It shows the current orientation of the view and assists in aligning objects with specific views.
- **Object Gizmo** : It allows you to move, scale or rotate objects using the mouse on any of the 3 axes.
- **Shading** : It refers to the way objects are displayed and lit in the 3D Viewport.
- **3D Cursor** : It is the position where new objects get created.
- **Wireframe Shading mode** : It helps to get a good view of the geometry and helps to view and select even those vertices/edges/faces that are not visible from the current view.